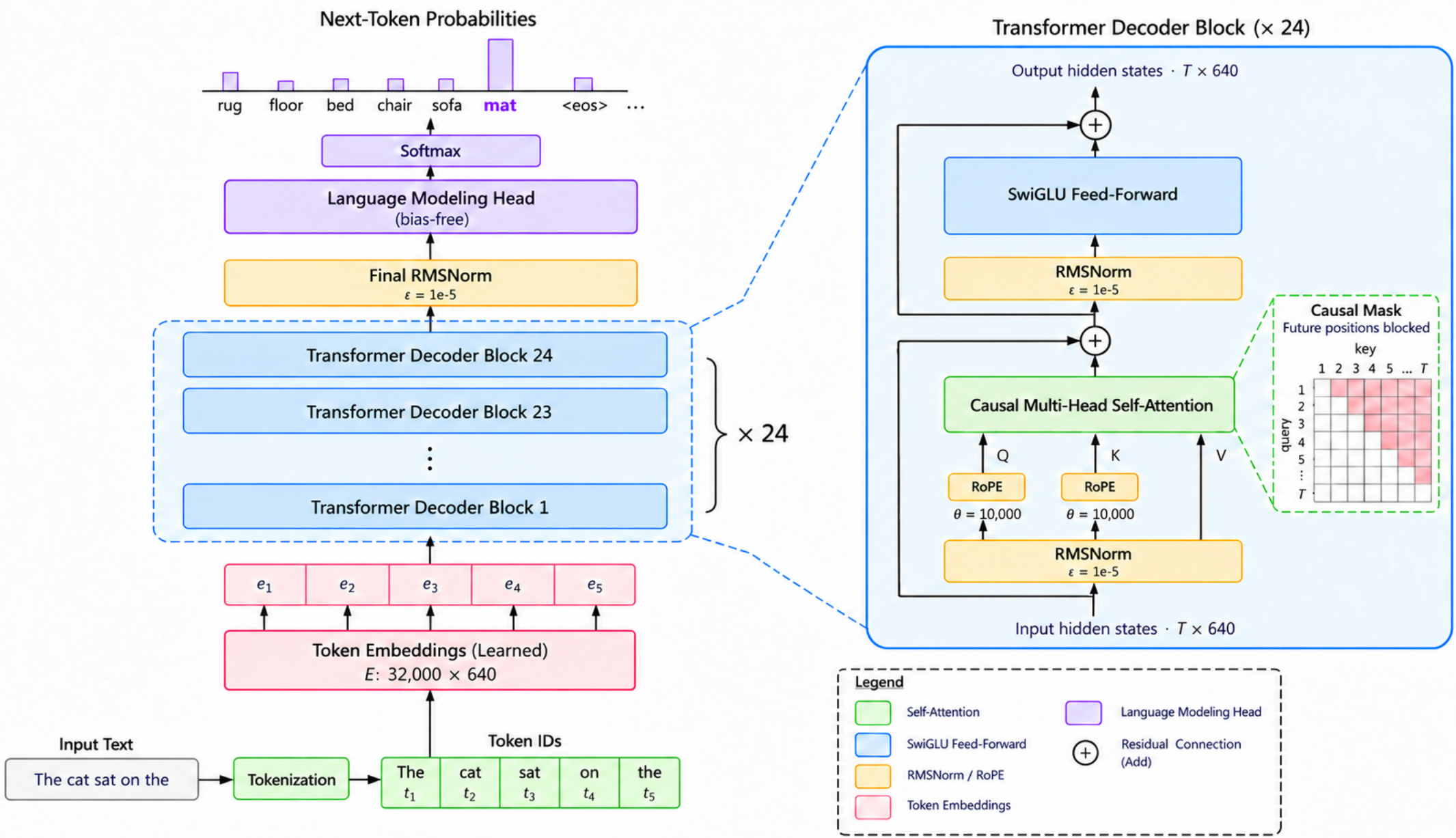

Next-Token Probabilities
rug
floor
bed
chair
sofa
mat
<eos>
...
Softmax
Language Modeling Head
(bias-free)
Final RMSNorm
ε = 1e-5
Transformer Decoder Block 24
Transformer Decoder Block 23
Transformer Decoder Block 1
× 24
e1
e2
e3
e4
e5
Token Embeddings (Learned)
E: 32,000 × 640
Input Text
The cat sat on the
Tokenization
Token IDs
The
t1
cat
t2
sat
t3
on
t4
the
t5
Transformer Decoder Block (× 24)
Output hidden states · T × 640
SwiGLU Feed-Forward
RMSNorm
ε = 1e-5
Causal Multi-Head Self-Attention
Q
K
V
RoPE
θ = 10,000
RoPE
θ = 10,000
RMSNorm
ε = 1e-5
Input hidden states · T × 640
Causal Mask
Future positions blocked
key
query
1 2 3 4 5 ... T
Legend
Self-Attention
SwiGLU Feed-Forward
RMSNorm / RoPE
Token Embeddings
Language Modeling Head
Residual Connection
(Add)